

Sainirnay S. Mantrala

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US Citizen

Education

University of Colorado, Boulder

Aug 2026 - May 2028

M.S. in Aerospace Engineering (Astrodynamics & Satellite Navigation Systems)

University of California, San Diego (GPA: 3.92)

Sep 2022 - June 2026

B.S. in Aerospace Engineering (Astrodynamics), Minor in Astrophysics

Professional Experience

Mission Design Intern — Intuitive Machines

June 2026 - Aug 2026

- Built Python-based mission analysis tools with web visualization interfaces to support lunar mission operations
- Created a SPICE-based ground station visibility tool generating AOS/LOS access windows with occultation, elevation, and range constraints; validated against GMAT line-of-sight predictions
- Designed deterministic spacecraft exclusion-zone avoidance algorithm, reducing toolset computational runtime by 46%

Aerospace Modeling and Simulation Intern — Maxar Technologies

June 2024 - Aug 2024

- Conducted high-fidelity physics-based Python modeling of space systems for sensor modeling, constellation scheduling, non-Earth imaging, and mission trajectory planning and presented clear, actionable insights to clients
- Refactored in-house mission planning toolsets to improve usability and expanded technical documentation
- Developed Cesium-based visualization toolset, reducing dependency on STK licenses and cutting software costs by 6%

Principal Investigator — NASA L'SPACE Proposal Writing and Evaluation Academy

May 2023 - Aug 2023

- Pitched proposal for a novel in-situ Martian cooling unit, securing interest from NASA Glenn Research Center
- Led an 11-member team in drafting a competitive NASA research proposal, authoring the Scientific Principles section
- Evaluated 10+ research proposals and provided expert feedback on technical feasibility as Primary Reviewer
- Researched cryogenic technologies, sustainable closed-loop coolant systems, and thermal engineering

Technical Experience

Independent Researcher — Dr. John Hwang, UCSD

Feb 2024 - May 2026

- Formulated high-fidelity trajectory optimization models for low-thrust orbital transfers using IPOPT, incorporating continuous-thrust dynamics and adaptive timestep control to achieve high accuracy and fast convergence
- Evaluated collocation and shooting-based formulations, comparing solution accuracy and convergence

LEO Orbit Determination — UCSD

Mar 2026 - June 2026

- Developed an 18-state nonlinear orbit-determination framework in MATLAB using iterative batch least-squares and sequential Kalman filtering with ground-based range and range-rate measurements
- Achieved centimeter-level range convergence in three batch iterations and characterized state uncertainty through covariance and measurement-sensitivity analysis

Orbital Mechanics Teaching Assistant — MAE 180, UCSD

Aug 2025 - Dec 2025

- Led discussion sections, office hour sessions, and grading to support undergraduate coursework in orbital mechanics
- Provided technical guidance on orbital transfers, orbit determination, and interplanetary trajectory design

Skills

- **Software** - Python, MATLAB, C++, Java, Git, Linux, SPICE, GMAT, Simulink, ANSYS, SolidWorks, FEMM
- **Analysis** - Astrodynamics, Optimization, State Estimation, Mission Design, Numerical Methods, Bayesian Statistics
- **Fabrication** - Additive Manufacturing, CNC Milling, Laser Cutting, Soldering, Potting